

SAVE THIS

 **AI Algorithms**

TOP 30 AI Algorithms Explained

Understanding their purpose and practical use.



Complete Overview

- | | | |
|--|--|---|
| 1 Linear Regression | 2 Logistic Regression | 3 Decision Tree |
| 4 Random Forest | 5 Support Vector Machine (SVM) | 6 K-Nearest Neighbors (K-NN) |
| 7 Naive Bayes | 8 Gradient Boosting | 9 AdaBoost |
| 10 XGBoost | 11 K-Means Clustering | 12 Hierarchical Clustering |
| 13 DBSCAN | 14 Principal Component Analysis (PCA) | 15 t-SNE |
| 16 Q-Learning | 17 SARSA | 18 Deep Q-Network (DQN) |
| 19 Policy Gradient | 20 Actor-Critic | 21 Artificial Neural Network (ANN) |
| 22 Convolutional Neural Network (CNN) | 23 Recurrent Neural Network (RNN) | 24 LSTM |
| 25 Transformer | 26 K-Means++ | 27 Autoencoders |
| 28 Isolation Forest | 29 Markov Decision Process (MDP) | 30 Genetic Algorithms |

Let's dive deep into each algorithm!

1

Linear Regression

WHAT IT DOES

Predicts a continuous number by finding the best straight-line relationship between input and output.

WHERE IT'S USED

- House price prediction
- Sales forecasting
- Stock price trend estimation
- Demand prediction

2

Logistic Regression

WHAT IT DOES

Predicts categories (usually yes/no) using probabilities.

WHERE IT'S USED

- Spam vs Not Spam email detection
- Disease detection (Positive/Negative)
- Loan approval (Approve/Reject)



3

Decision Tree

WHAT IT DOES

Makes decisions using a tree-like structure of questions (if-else conditions).

WHERE IT'S USED

- Loan approval systems
- Customer churn prediction
- Medical diagnosis
- Risk analysis

4

Random Forest

WHAT IT DOES

Builds many decision trees and combines their results.

WHERE IT'S USED

- Fraud detection
- Credit scoring
- Recommendation systems
- Medical predictions



5

Support Vector Machine (SVM)

WHAT IT DOES

Finds the best boundary to separate different classes.

WHERE IT'S USED

- Face recognition
- Text classification
- Bioinformatics
- Image classification

6

K-Nearest Neighbors (K-NN)

WHAT IT DOES

Classifies a data point based on its closest neighbors.

WHERE IT'S USED

- Recommendation systems
- Pattern recognition
- Image recognition
- Simple classification tasks



7

Naive Bayes

WHAT IT DOES

Uses probability to predict categories. Assumes features are independent.

WHERE IT'S USED

- Spam detection
- Sentiment analysis
- Document classification
- News categorization

8

Gradient Boosting

WHAT IT DOES

Builds models step-by-step. Each new model fixes previous mistakes.

WHERE IT'S USED

- Ranking systems (Google search ranking)
- Fraud detection
- Customer behavior prediction



9

AdaBoost

WHAT IT DOES

Combines many weak models into one strong model.

WHERE IT'S USED

- Face detection
- Binary classification problems
- Real-time detection systems

10

XGBoost

WHAT IT DOES

A highly optimized and fast version of gradient boosting.

WHERE IT'S USED

- Machine learning competitions
- Business analytics
- Credit scoring
- Sales prediction



11

K-Means Clustering

WHAT IT DOES

Groups similar data into clusters.

WHERE IT'S USED

- Customer segmentation
- Market analysis
- Image compression

12

Hierarchical Clustering

WHAT IT DOES

Builds clusters step-by-step into a tree-like structure.

WHERE IT'S USED

- Genetic data grouping
- Document clustering
- Social network analysis



13

DBSCAN

WHAT IT DOES

Groups dense areas and detects outliers.

WHERE IT'S USED

- Fraud detection
- Geographic data analysis
- Anomaly detection

14

Principal Component Analysis (PCA)

WHAT IT DOES

Reduces number of features while keeping important information.

WHERE IT'S USED

- Image compression
- Noise reduction
- Feature reduction before ML training



15

t-SNE

WHAT IT DOES

Helps visualize complex data in 2D or 3D.

WHERE IT'S USED

- Visualizing embeddings
- Understanding clusters
- Data exploration

16

Q-Learning

WHAT IT DOES

Learns best action in a situation to maximize long-term reward.

WHERE IT'S USED

- Robotics
- Game AI
- Autonomous systems



17

SARSA

WHAT IT DOES

Similar to Q-learning but updates based on current action policy.

WHERE IT'S USED

- Navigation systems
- Game agents

18

Deep Q-Network (DQN)

WHAT IT DOES

Uses neural networks with Q-learning.

WHERE IT'S USED

- Playing video games (Atari)
- Self-driving simulations



19

Policy Gradient

WHAT IT DOES

Directly learns best action strategy.

WHERE IT'S USED

- Robotics
- Continuous action problems

20

Actor-Critic

WHAT IT DOES

Actor decides action, critic evaluates action.

WHERE IT'S USED

- Advanced robotics
- Complex game AI
- Real-time decision systems



21

Artificial Neural Network (ANN)

WHAT IT DOES

Basic neural network inspired by human brain.

WHERE IT'S USED

- Pattern recognition
- Prediction problems

22

Convolutional Neural Network (CNN)

WHAT IT DOES

Specialized for image data.

WHERE IT'S USED

- Face recognition
- Medical image analysis
- Self-driving cars



23

Recurrent Neural Network (RNN)

WHAT IT DOES

Processes sequential data.

WHERE IT'S USED

- Text prediction
- Speech recognition
- Time-series forecasting

24

LSTM

WHAT IT DOES

Advanced RNN that remembers long-term dependencies.

WHERE IT'S USED

- Language translation
- Chatbots
- Stock prediction



25

Transformer

WHAT IT DOES

Uses attention mechanism to process entire sequences at once.

WHERE IT'S USED

- ChatGPT
- BERT
- Machine translation
- Text summarization

26

K-Means++

WHAT IT DOES

Improved version of K-Means with better starting points.

WHERE IT'S USED

- Clustering problems for better accuracy



27

Autoencoders

WHAT IT DOES

Neural networks that compress and reconstruct data.

WHERE IT'S USED

- Noise removal
- Feature learning
- Image compression

28

Isolation Forest

WHAT IT DOES

Detects unusual data points (anomalies).

WHERE IT'S USED

- Fraud detection
- Intrusion detection
- Financial monitoring



29

Markov Decision Process (MDP)

WHAT IT DOES

Mathematical framework for decision-making.

WHERE IT'S USED

- Reinforcement learning
- Robotics
- Planning systems

30

Genetic Algorithms

WHAT IT DOES

Inspired by evolution (selection, mutation, crossover).

WHERE IT'S USED

- Optimization problems
- Scheduling
- Route planning
- Engineering design





Was this helpful?
Follow me for more



@ai.with.akki